

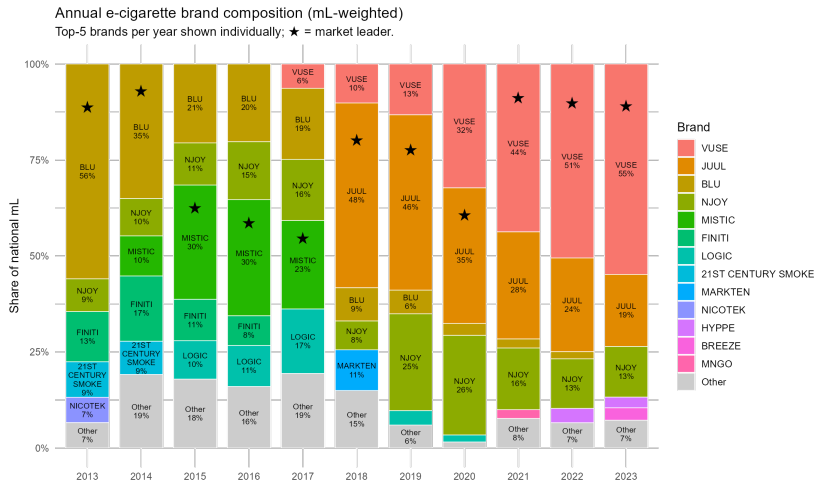
A Game of Leapfrog: Regulation and Market Movements in the U.S. E-Cigarette Market

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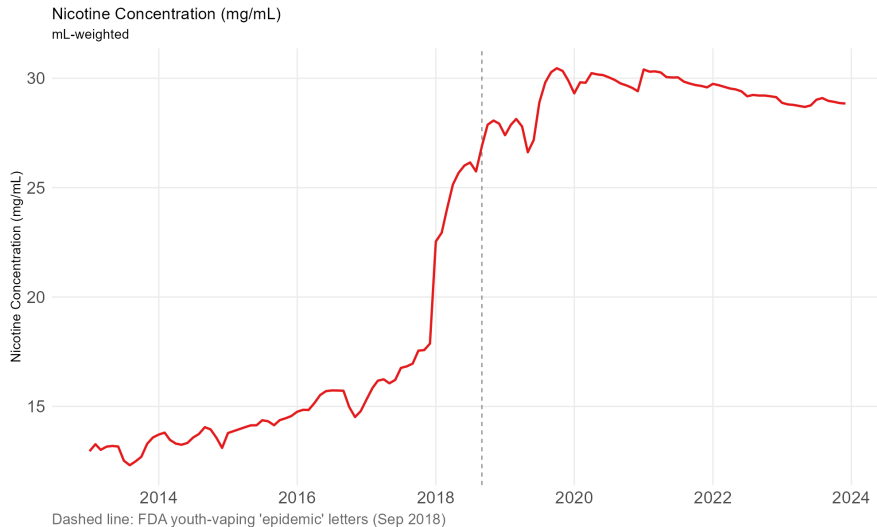
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Motivation



- Small- N , leadership turns over, “Other” (the fringe) is non-negligible

Nicotine concentration has risen



FDA interventions 2018+

- 2018 FDA enforcement begins — youth-vaping “epidemic” declared; flavor-restriction plan announced (Nov)
- 2019 EVALI health scare (fall); JUUL pulls fruit/mint flavors (Nov)
- 2020 Flavored-cartridge enforcement takes effect (Feb); market authorization requirement (Sep)
- 2022 Synthetic-nicotine loophole closed; FDA denies JUUL’s marketing authorization (Jun)

Counterfactuals

Absent this escalation, how would:

- Nicotine concentration (mg/mL) have evolved?
- Market leadership have evolved?

The plan

Today

Static pricing, **exogenous** quality state, dynamic entry & exit.

Future

Adds **endogenous** investment (binary invest/don't-invest), FDA regulation entering through a firm-and-time-varying investment cost.

- Kilts Nielsen Retail Scanner Data
- Weekly store sales, UPC (barcode) level, 2013–2023
- Aggregated to firm $\times$ month

Environment

Players (period t , monthly):

- Up to $\bar{N} = 5$ incumbent firms, $i = 1, \dots, n_t \leq \bar{N}$
- One explicit competitive fringe F (aggregated)
- At most one potential entrant per period

State: $\omega_{i,t} = (a_{i,t}, \xi_{i,t})$

- $a_{i,t}$: accumulated-quality index — **observed** data. Product-unweighted average nicotine content (mg/mL) across firm i 's active products that month
- $\xi_{i,t}$: idiosyncratic residual — **not** observed directly; recovered from the demand estimation
- Both evolve exogenously via first order Markov

$a_{i,t}$ evolution by firm $\rightarrow$

Each period: states realize $\rightarrow$ static pricing subgame $\rightarrow$ profits realize $\rightarrow$ exit/entry decided $\rightarrow$ states evolve.

Nested logit: incumbents (i) and the fringe (F) share one e-cigarette nest against a degenerate outside-option nest.

$$\delta_{i,t} = \beta_a a_{i,t} + \xi_{i,t} - \alpha p_{i,t}$$

- $a_{i,t}$: observed quality (nicotine concentration, mg / mL)
- $\xi_{i,t}$: exogenous residual
- $p_{i,t}$: simple mean of UPC-month prices (real \$/mL), each computed as real revenue / mL sold.

Estimated via Berry (1994). [IV/2SLS →](#)

Profit-measurement caveat: this price times firm mL need not equal observed revenue.

Static pricing subgame

$$\pi_i(\omega_t) = \max_{p_{i,t}} (p_{i,t} - c(\omega_{i,t})) q_i(p_t, \omega_t) - FC$$

Bertrand–Nash equilibrium price vector, solved period-by-period given the vector of all firms' states, ω_t .

$q_{i,t} = M_t \cdot s_i(p_t, \omega_t)$: quantity = market size $\times$ predicted share.

Dynamic entry and exit

Incumbent's value function:

$$V_i(\omega_{i,t}, \omega_{-i,t}) = \pi_i^*(\omega_t) + \max \left\{ \underbrace{\phi_i}_{\text{exit: scrap value}}, \underbrace{\beta \mathbb{E}[V_i(\omega_{i,t+1}, \omega_{-i,t+1}) \mid \omega_t]}_{\text{continue}} \right\}$$

$\beta \in (0, 1)$: discount factor.

Entrant's problem: enter iff

$$\beta \mathbb{E}[V_e(\omega_{e,0}, \omega_{-e,t+1}) \mid \omega_t] - \underbrace{x_e}_{\text{sunk entry cost}} \geq 0$$

Equilibrium: a **Markov Perfect Equilibrium (MPE)** — a fixed point in the exit/entry cutoff rules and the value function — computed via Pakes–McGuire (1994) iteration.

The fringe

- Model explicitly due to non-negligible market share, and to avoid curse of dimensionality
- Affects incumbents' pricing decisions via entering nested logit demand
- Fringe always exists, therefore no entry/exit decision
- Its state $a_{F,t}$ is constructed directly from the data, the same way as $a_{i,t}$

Looking forward

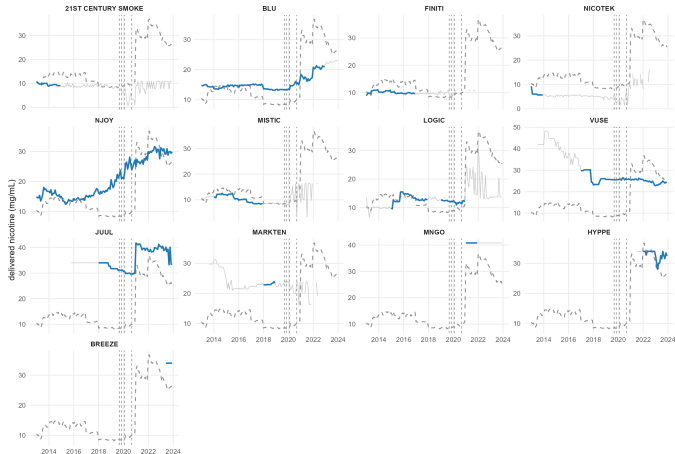
- ① **Endogenous** $a_{i,t}$. Firms choose to invest in quality. Leads to large state space.
 $MPE \rightarrow EBE$ Why? $\rightarrow$
- ② **Period length.** Monthly (current) vs. quarterly vs. annual — richer regulatory-event timing and more transitions to estimate from, against computational burden.
- ③ **The fringe.** Market share endogenous.
- ④ $\bar{N} = 5$ **at capacity.** When the roster is full and a firm wants to enter, does that require a same-period incumbent exit?

Questions

$a_{i,t}$ evolution by firm

$a_{i,t}$: tracked firms, one panel each

Blue/bold = that firm's actual top-5 (incumbent) years; thin grey = tracked but not top-5 that year; dashed grey = fringe (pooled), shown for reference in every panel



Events: 1 EVALI scare (fall 2019); 2 JUUL pulls fruitmint (Nov 2019);
3 FDA flavored-cartridge enforcement (Feb 2020); 4 PMTA deadline (Sep 2020)

Berry (1994) inversion

With aggregate share data, σ , β_a , and α are recovered from a single linear IV/2SLS regression.

$$\underbrace{\ln(s_{i,t}) - \ln(s_{0,t})}_{\text{log share ratio}} = \underbrace{\sigma \ln(s_{i,t}/s_{g,t})}_{\text{within-nest share}} + \beta_a a_{i,t} - \alpha p_{i,t} + \xi_{i,t}$$

- $s_{i,t}$, $s_{0,t}$, $s_{g,t}$: firm, outside-option, and nest shares — fringe volume enters the denominators here, but the fringe is not its own observation
- Estimated on incumbents only; $\xi_{i,t}$ is the regression residual

Pakes–McGuire (1994) iteration

Computes a MPE in a dynamic oligopoly model with a large, discrete state space.

- 1 Guess a value function for each firm.
- 2 Given that guess, solve each firm's optimal exit and entry thresholds from the Bellman equation.
- 3 Update the value function by iterating the Bellman equation forward.
- 4 Repeat until both the policies and the value function converge.

A fixed point in policies *and* values jointly — not solved in closed form. Feasible here because payoffs and transitions are anonymous (depend on the state vector, not on firm identity).

Why Experience-Based Equilibrium, eventually

Introducing endogenous investment leads to a large state space:

- Up to $\bar{N} = 5$ firms $\times$ 2-dimensional state = up to 10 dimensions
- A firm's best response must be optimal against rivals' *entire* state vector

Fershtman & Pakes (2012), Experience-Based Equilibrium

Relaxes consistency to states likely to happen, and lets firms condition on a coarser summary of rivals rather than the full state vector.